1-Task Title: Bezout's Co-efficient:

Code:

Bezouts_coefficient.cpp

```
#include<bits/stdc++.h>
using namespace std;
#define newl cout<<endl

int gcd;
vector<int>quotient_gcd(int a,int b, vector<int>q)
{
   if(b==0)
   {
      gcd=a;
      return q;
   }
   if((a/b)!=0)
   {
      q.push_back(a/b);
   }
   return quotient_gcd(b,a%b,q);
}

map<int,int>Bezouts_coefficient(int a,int b)
{
   vector<int>s={1,0},t={0,1},q;
   q=quotient_gcd(a,b,q);
   for(int i=2;i<=q.size();i++)
   {
      s.push_back(s[i-2]-q[i-2]*s[i-1]);
      t.push_back(t[i-2]-q[i-2]*t[i-1]);
   }
   map<int,int>res;
   res[max(a,b)]=s.back();
   res[min(a,b)]=t.back();
   return res;
}
```

```
#include"Bezouts_coefficient.cpp"
int main()
{
   cout<<"Enter 2 numbers-> ";
   int a,b;
   cin>>a>>b;
   map<int,int>res=Bezouts_coefficient(a,b);
   newl;
   cout<<"Bezouts Coefficient of "<<a<<" is "<<res[a];
   newl;
   cout<<"Bezouts Coefficient of "<<b<<" is "<<res[b];
   newl;
}
```

Output:

```
Enter 2 numbers-> 252 198

Bezouts Coefficient of 252 is 4
Bezouts Coefficient of 198 is -5
PS F:\RUET CSE LAB\Discrete Mathematics 2102\8. 20-04-2024>
```

Discussion:

Here I have used Back Extended Euclidean Algorithm to calculate the Bezuot's Coefficients. When the GCD of 2 values is 1 then the Bezout's Coefficients are the inverse value to each other's. These values are necessary for calculating the simultaneous congruence relation like Chinese Remainder Theorem.

2-Task Title: Chinese Remainder Theorem:

Code:

```
#include"Bezouts_coefficient.cpp"
int main()
{
   cout<<"Enter Total Congruence Relations (a,m): ";
   int n;
   cin>>n;
   ll M=1;
   int x=0;
   vector<pair<int,int>>CR;
   for(int i=1;i<=n;i++)
   {
      int a,m;
      cin>>a>>m;
      CR.push_back({a,m});
      M*=m;
   }
   for(int i=0;i<n;i++)
   {
      ll mk=M/CR[i].second;
      map<int,int>inverse=Bezouts_coefficient(CR[i].second,mk);
      x+=(CR[i].first*mk*inverse[mk]);
   }
   if(x%M<0)
   {
      x=(x%M)+M;
   }
   cout<<"The value of x is ";
   cout<<(x%M);
}
```

Output:

```
Enter Total Congruence Relations (a,m): 3
7 9
0 4
4 5
The value of x is 124
PS F:\RUET CSE LAB\Discrete Mathematics 2102\8. 20-04-2024>
```

Discussion:

Chinese Remainder Theorem is a process to calculate the simultaneous congruence relation where the moduli are pairly relatively prime. Here I have used Bezout's theorem to calculate the inverse. After that I put them in the equation of the theorem to calculate the minimum positive number which satisfies the congruence relations.